

User & Job: Hoc vien can hoi lai bai ma khong lac nguon

Bang chung: chatlog co 2,522 dong, gom 1,261 luot student va 1,261 luot tutor.

Core JTBD: Khi dang on slide/transcript VLearn, hoc vien muon hoi bang ngon ngu tu nhien va tim dung phan bai hoc, de hieu nhanh ma khong phai tu keo tung trang PDF.

Log co nhieu cau phu thuoc ngu canh: "tom tat noi dung chinh trong slide nay", "giai thich 4 chien luoc", "cach xu ly ngu canh".

Nguon: data/vlearn-pack/chatlog/chat_history_anonymized_for_hackathon.csv

Vi sao chọn VLearn Knowledge Tutor?

So sanh 4 y tuong:

1. Tutor hoi dap theo slide/transcript - Impact cao - CHON.
2. Chatbot tong quat cho hoc vien - qua rong, de bia - LOAI.
3. Auto-generate quiz/dap an - rui ro lam bai ho - LOAI o moc nay.
4. Search-only PDF navigator - huu ich nhung chua du quyet dinh AI - GIU lam phu.

Ly do chon: co du lieu that, do duoc bang eval, demo duoc trong 5 phut.

Nguon: spec.md section 5; eval/test_cases.json

Giai phap: 2 tool cho chatbot hoc tap

AI quyet dinh: hoi bai hoc, tro chuyen thuong, hoi lai hay tu choi.

Tool 1 - lesson_search_tool: tim trong slide/transcript, tra ve day, page, citation, text.

Tool 2 - web_search_tool: dung RapidAPI web search khi cau ngoai slide can thong tin ngoai.

Demo live case chuan: "Agent trong Day1 la gi?" -> tra loi co nguon.

Demo live case kho: "Day1 co nhac GPT-9 khong?" -> noi khong co can cu, khong bia.

Nguon: tools/lessonSearchTool.js; tools/webSearchTool.js; server.js

Ket qua eval: dat quality bar

Quality bar: $\geq 80\%$ test case dat va khong duoc bia nguon du chi mot lan.

Bo test hien co: 30 case.

- 4 case thong tin khong co trong tai lieu.
- 4 case cau mo ho/thieu ngu canh.
- 4 case yeu cau khong duoc phep.
- 5 case tra loi sai gay hau qua that.
- 13 case normal chat + lesson answer + UI navigation.

Lan chay gan nhat: 30/30 dat = PASS 24, PASS_REVIEWED 2, PASS_CODE_CHECK 4.

Nguon: eval/results_latest_summary.json; eval/results_latest.csv

Validation: phan hoi lam thay doi san pham

Quote 1: "vi du toi tim agent nhung slide k nhay toi phan agent".

-> Sua click search/citation de goi jumpToSource va reload PDF dung page.

Quote 2: "bay gio trong chatbot toi khong muon nhay slide nua".

-> Bo auto-jump trong chatbot; user tu bam citation/search result.

Quote 3: "chatbot dg k doc dc het slide khi tom tat day1".

-> Voi intent tom tat Day, gui toan bo slide cua Day do thay vi top 3 nguon.

Nguon: validation/feedback_log.md

Neu co them 1 tuan

Uu tien 1: chay validation voi it nhat 5 nguoi ngoai nhom, ghi quote that.

Uu tien 2: cau hinh mot RapidAPI web search cu the de web_search_tool chay end-to-end.

Uu tien 3: highlight doan lien quan trong PDF/slide, khong chi nhay toi page.

Bai hoc lon nhat: AI tutor dang tin khong phai vi tra loi nhieu,

ma vi biet dung nguon, hoi lai va tu choi dung luc.

Nguon: spec.md section 9, eval/results_latest_summary.json, validation/feedback_log.md